# 📘 Project Description: *Wave Enterprise*

## 1. Description

*Wave Enterprise* is an e-commerce CMS web application designed to serve as a digital furniture catalog and online storefront. It provides customers with a smooth shopping experience while giving administrators powerful tools to manage products and orders. The system integrates traditional retail practices with modern digital commerce by enabling direct WhatsApp communication for order fulfillment.

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## 2. Features

- **Customer Features**
  - Account creation and management
  - Automatic age and gender detection from ID
  - Product catalog browsing
  - Add products to cart and wishlist
  - Place orders with direct WhatsApp integration
- **Admin Features**
  - Admin account creation and management
  - Full product management (add, edit, delete, update)
  - Inventory oversight
  - Monitor customer orders and accounts

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## 3. Pages

- **Customer-facing Pages**
  - Home page (featured products, promotions)
  - Product catalog (browse furniture categories)
  - Product detail page (images, description, price)
  - Cart page (selected items, checkout option)
  - Wishlist page
  - Account page (profile, ID-based details, order history)
- **Admin-facing Pages**
  - Admin dashboard
  - Product management (CRUD operations)
  - Order management
  - User management

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## 4. Timeline

- **Phase 1: Planning & Requirements (2 weeks)**
- **Phase 2: Development (6–8 weeks)**
- **Phase 3: Testing & Deployment (2–3 weeks)**
- **Phase 4: Future Enhancements (Ongoing)**

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## 5. Roles & Role Actions

- **Customer**
  - Register and manage account
  - Browse catalog, add to cart/wishlist
  - Place orders via WhatsApp
- **Admin**
  - Create and manage admin accounts
  - Manage product listings and inventory
  - Oversee customer orders and accounts

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## 6. Limitations

- No integrated payment gateway (orders handled via WhatsApp chat)
- Limited personalization beyond ID-based age/gender detection
- Wishlist and cart features are basic (no advanced recommendation engine yet)
- Scalability may require optimization for larger product catalogs
- Voluntary project scope means some advanced features are deferred for future updates

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## 7. Target Audience

- Furniture retailers seeking a digital storefront
- Customers looking for convenient online shopping experiences
- Small businesses wanting a low-cost e-commerce solution

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## 8. Objectives

- Provide a user-friendly furniture catalog system
- Enable direct customer-to-seller communication via WhatsApp
- Simplify product and inventory management for admins
- Lay the foundation for future integration of payment gateways and analytics

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## 9. Technology Stack (High-Level Overview)

- **Frontend:** Thymeleaf, HTML, Bootstrap 5.3.3, JavaScript
- **Backend:** Spring Boot 3 (Java)
- **Database:** MariaDB

- **Web Server:** Tomcat
- **IDE:** Visual Studio Code

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## 10. Future Enhancements

- Integration of secure online payment gateways
- Recommendation engine for personalized shopping
- Advanced analytics dashboard for admins
- Mobile app version for Android/iOS
- Multi-language and multi-currency support

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## 11. Project Impact

- Digitizes traditional furniture retail for *Wave Enterprise*
- Improves customer engagement and accessibility
- Provides scalable foundation for future e-commerce growth

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## 12. Security & Privacy Considerations

- Secure customer authentication and account management
- Data encryption for sensitive information
- Role-based access control for admins vs customers
- Compliance with local data protection regulations

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## 13. Competitive Advantage

- Direct WhatsApp order integration (unique compared to many e-commerce platforms)
- Lightweight CMS tailored for small-to-medium businesses
- Simple, intuitive design for both customers and admins

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## 14. Risks & Challenges

- Reliance on WhatsApp for order communication may limit scalability
- Absence of integrated payment gateway could reduce customer trust
- Future expansion may require migration to more advanced hosting solutions